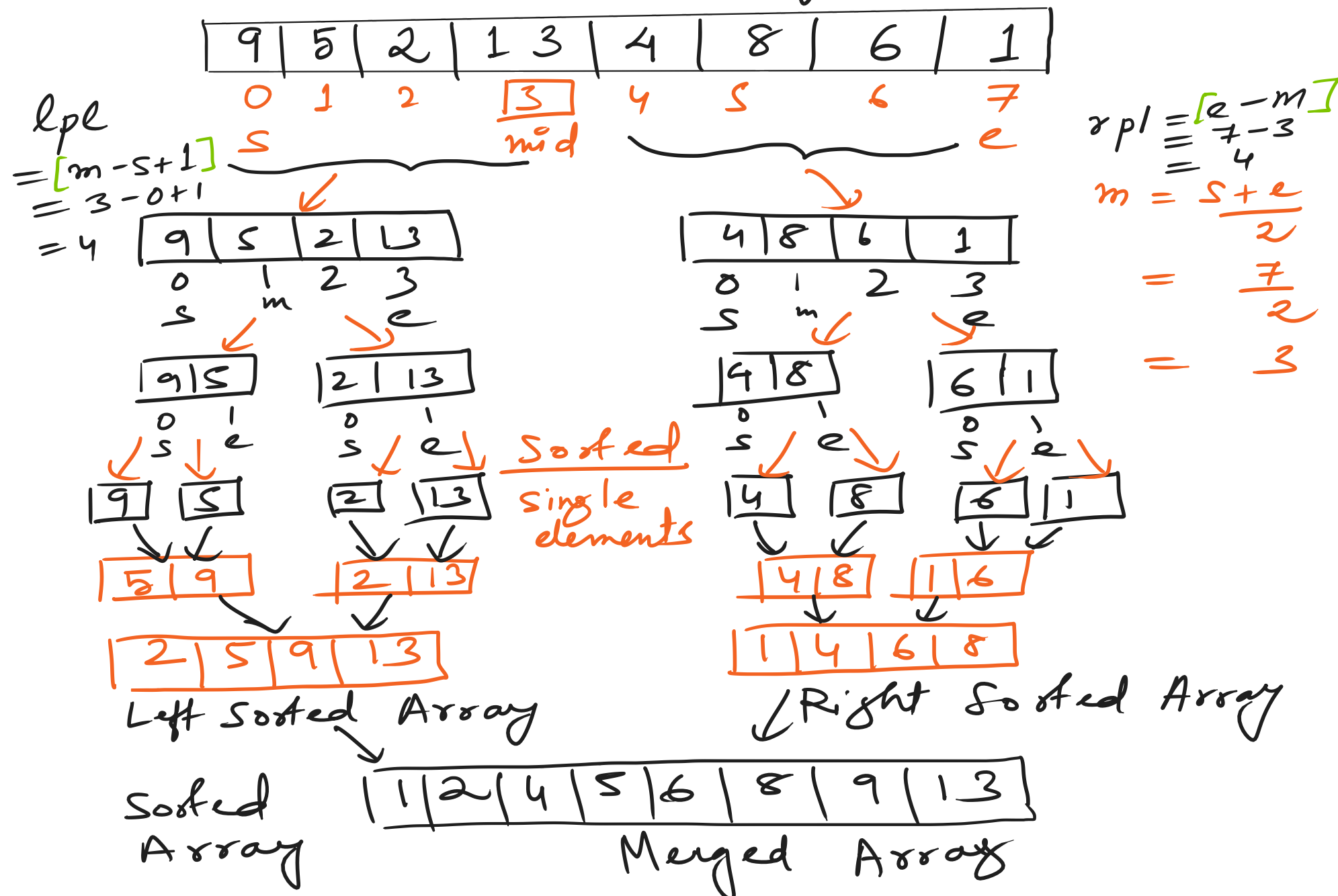


Merge Sort Algorithm Divide & Conquer Algos

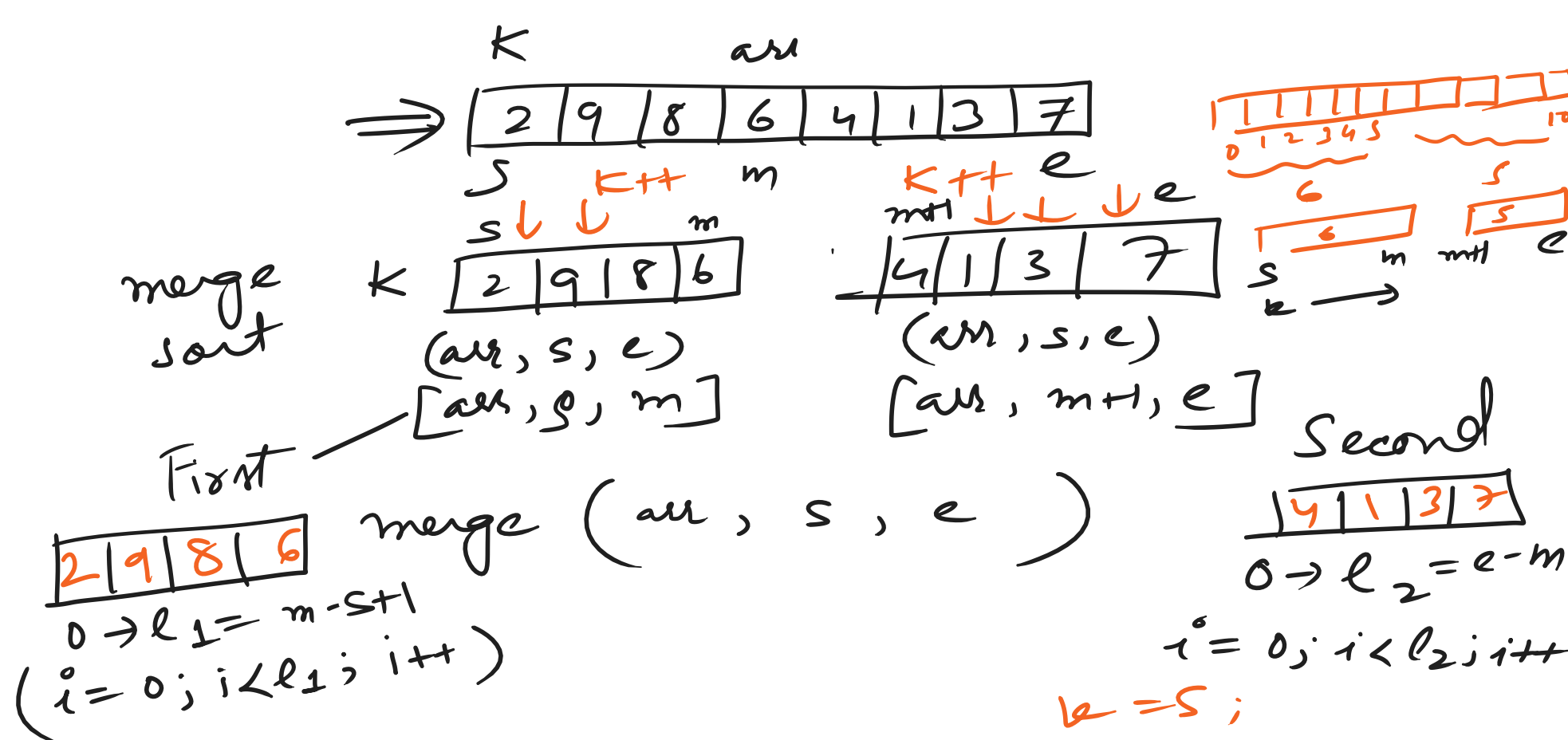
The array is divided into parts/halves as long as we don't get a single element $\rightarrow$ which is considered sorted in itself. Then single elements are compared and two arrays are created & then merged using "recursion".



Time Complexity:

- ① Dividing the array: $O[\log N]$
- ② Merging two sorted arrays: $O[N]$

Total Time Complexity = $N \times \log N$
 $= O[N \log N]$



Non-comparison algorithms:

- ① Count Sort
- ② Radix Sort

Count Sort Algorithm:

Key Points:

- $\rightarrow$ Single digit whole numbers $\rightarrow 0-9$
- $\rightarrow$ Non-comparison algorithm

Steps:

1. Find the max
 $\text{max} \rightarrow K = 6$
2. Create a count array
 $[0-K] \Rightarrow 0-6$
3. Calculate the no. of occurrences of each element
 $L \rightarrow R$
4. Calculate cumulative count
 $C = C + P$
5. O/p array = size(inp)
6. Start from end
7. Copy the content back to the input array.

Time Complexity
 $O(n+k) \Rightarrow \text{Linear}$

